

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Technology (CSE)

QUIZ #2 (SET-A)**DURATION: 15 MINUTES****SUMMER SEMESTER, 2024-2025****FULL MARKS: 15**

CSE 4621: Machine Learning

Programmable calculators are not allowed.

Answer **all 5 (five)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

Student ID:

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1. How can logistic regression produce a nonlinear decision boundary in the original input space? 3
- A. By replacing gradient descent with Newton's method (CO1)
- B. By adding polynomial features** (PO1)
- C. By adding the sigmoid function
- D. By setting all weights equal

Solution:

Logistic regression remains linear in its parameters, but nonlinear feature transformations such as: x_1^2 , x_2^2 , or x_1x_2 can create nonlinear boundaries in the original feature space.

2. Why the does the universal approximation theorem not guarantee useful learning from finite data? 3
- A. **Approximation capacity does not mean successful optimization** (CO1)
- B. Approximation capacity is limited by the feature space (PO1)
- C. The theorem prohibits linear functions with continuous output
- D. The theorem does not allow the model to rescale and shift normalized activations

Solution:

A network may have enough capacity to represent the target function, but training may not find the needed parameters, and fitting observed data does not ensure good performance on unseen examples.

3. How does batch normalization behave differently during training and inference? 3
- A. The activation function changes at inference, making the gradients smoother (CO1)
- B. Gradients are computed during inference and then saved in memory for further training (PO1)
- C. Inference is not done in mini-batches, only sequentially
- D. Training uses mini-batch statistics, while inference uses stored estimates**

Solution:

During training, batch mean and variance are used. During inference, running estimates provide stable deterministic normalization.

4. In gradient descent, a learning rate that is excessively large is most likely to cause:

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- A. Perfect generalization
- B. All parameters become zero
- C. Oscillation or divergence**
- D. The loss to become convex

(CO1)

(PO1)

Solution:

A large learning rate can overshoot low-loss regions repeatedly and may cause the optimization process to diverge.

5. Logistic regression and conventional neural-network classifiers typically assume that each input instance belongs to exactly one class. How would you design a classification system for cases in which a single instance may belong to multiple classes simultaneously?

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(CO2)

(PO1)

Solution:

Use a multi-label classification model with one sigmoid output neuron per class, exactly like regular neural networks. Train the model using binary cross-entropy loss for each output, and assign every class whose predicted probability exceeds a chosen threshold.

Rubric:

- 1 mark for the correct formulation of ideas.
- 2 mark for a valid explanation.